Step5 :- Merge branch and Resolve Conflict.
Bash

```
git checkout main
git merge f1
```

if conflict occurs:

1. open conflicted file
2. Resolve manually.
3. Commit again.

Bash

```
git add first.txt
git commit -m "conflict Resolved"
```

Step6 :- Push Changes
Bash

```
git push origin main.
```

→ Important commands.

* git branch :- Creates a new branch
* git Checkout :- Switches branches
* git (branch) merge :- Merges branches
* git diff :- shows file differences
* fork :- Creates copy of repository on GitHub.

Observed :- Forking, branching, merging, and conflict resolution were sucessfully performed using GitHub

Aim :- To demonstrate how to create a project in a remote repository and apply fork, branch, merge, conflict, diff, and pull request concepts using GitHub.

Procedure:-

step1 :- Fork a repository.

1. Login to GitHub.
2. Search for an Existing repository (Example: Add Two numbers)
3. Click on fork.
4. Rename the repository if required.

step2 :- Clone Forked Repolitory

Bash

```
git clone {forked-repo-url}
cd {repository-name}
```

step3 :- Create and switch Branch.

Bash

```
git branch f1
git checkout f1
```

step4 :- Modify file and commit.

Bash

```
nano first.txt
git add first.txt
git commit -m "updated content in f1"
```

Step 5 :- Build Configuration
Choose Execute Windows Batch Command :
Bat

```
Javac MyFile.java
java MyFile
```

Step6: Build Project
Click "Build Now"

Result:
The Java project was successfully executed automatically using Jenkins CI/CD pipeline.

Installation of jenkins and setup.

Step1: Download Jenkins

* Go to the official website jenkins
* Click download and deploy
* Choose Windows (.msi installer)

Step2: Install jenkins on Windows.

i) Double-click the downloaded .msi file.
ii) click next
iii) choose installation directory.
iv) choose Port number (default : 8080).

Aim :- To demonstrate the integration of GitHub repository with Jenkins to automate project execution using CI/CD pipeline.

Procedure :-

step1 :- Install Java (JDK 21).
1. Download Java JDK 21 from Oracle website.
2. Install JDK.
3. Set Java path Environment variables.

step2 :- Create a java project.
1. Open Vs code.
2. Create MyFile.java.
3. write Simple java program.
4. Save file.

step3 :- Push Java Project on GitHub
Bash
```
git add MyFile.java
git commit -m "jenkins java file"
git push
```

step4 : Jenkins Configuration
1. Open Jenkins.
2. Create New Job → Freestyle Project.
3. Select Git as source code Management.
4. Paste GitHub repository url.
5. Select branch as main.

Experiment No. ..... Date ..... Page No.

Name of Experiment .....

step6: add, commit and push the .java file into remote (git) repository.

$ git add simple.java
$ git commit -m " My java file"
$ push.

-> Execution of code using jenkins.

step1: login to jenkins using username and password.

step2: click on "new item" and name the item as per convienience.

step3: click on "free-style Project" and ok.

step4: Source code management -> Git
-> Repository URL : " https://github.com/vve23cseaiml0101/progs.git.

step5: chane branch from 'master' to 'main

Step6: In trigger; select github hook trigger for

Experiment No. ..... Date ..... Page No.

Name of Experiment .....

→ Creation of new repository and dumping a java file into the github repository.

step1: ~~clone~~ open browser and log in to github repository.

Step2: Create a new repository with name "prog5".

Step3: In git bash clone the 'prog5" repo into local repository. using,

$ git clone {https://github.com/vvce23cseaiml0101-sketch/prog5.git}

step4: do all required configuration using

$ git ~~glob~~ config --global user.name {name}
$ git config --global user.email {email}

step5: open Vs studio code and create and store a simple .java file into local repository.

Experiment No. ..... Date ..... Page No.
Name of Experiment .....

v) Select Run service as Local System
vi) Click instace.
vii) Click Finish.

Step3 : Unlock Jenkins

i) Open browser:
https://localhost/8080.
Create.
i) Username
ii) password → confirm password
iii) name
iv) Email.
note:
Before password setup. system asks for administrator password. for that

i) copy the address-link displayed on the screen
ii) copy and paste it on another tab in ctrl-space.
iii) A random string will be generate and can be used as password.

ii) Click save and continue
iii) Confirm Jenkins URL (http://localhost:8080)
iv) Click Start using Jenkins

Experiment No. ..... Date ..... Page No.

Name of Experiment .....

Create a Maven project with all dependencies required for the application in CI/CD pipeline.

Step1:- open eclipse IDE platform.
Step2:- New → maven project.
Step3:- Select 'org. apache. maven. quickstart'
Step4:- {Groupid - exp, Artifact id - Exp.}
Step5:- finish.
Step6:- type 'Y' at console to "Build Success".
Step7:- open pom.xml file and minimize the window.
Step8:- open browser → maven repository → 1.1 version → copy the dependencies
Step9:- replace the dependencies from (35-37) lines in pom.xml file and save.
Step10:- Create a folder 'jsonfile' in Exp and a file in jsonfile, named exp.json.

Step11:- Create a new class example1.ja under "Src/test/java".

Experiment No. ..... Date ..... Page No.

Name of Experiment .....

Step7: build setup.
click and select " Execution windows batch command".

Step7: In command space type.

javac {file.name} .java

Step8:- Select build now; A green signal of success state will be appeared.

Experiment No. ..... Date ..... Page No.

Name of Experiment .....

exp.json

Example1.java.

```
public class example1 {
  public static void main (String[] args)
{
 JSONParser jsonparser = new JSONParser();
FileReader filereader = new FileReader("
                  .\\jsonfile \\emp.json");
object obj = jsonparser.parse (filereader);
JSONObject empjob = (JSONObject) obj;
String fname = (String) empjob.get ("firstname");
String lname = (String) empjob.get ("lastname");

System.out.println("Firstname" + fname);
System.out.println(" Last Name" + lname);
}
}
```